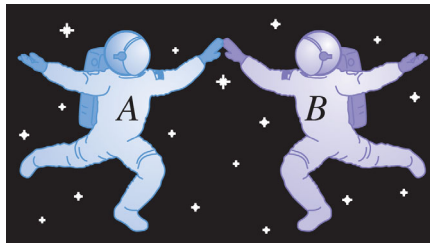


Lecture 5 – Linear momentum and impulse, collisions

- A/Prof Tay Seng Chuan
Email: pvotaysc@nus.edu.sg

When two objects collide, how do their velocity and energy change during their contact time and after their separation? Is there a price to be paid for remaining together after an event of collision? In this lecture we consider the answers in terms of the concept of momentum.



A few slides in this chapter come from Chapter-8 of Young's textbook.
This PPT file will well define the scope of your tests.

1

Acceleration and velocity – a review

Consider a particle of mass m moving in a 3D space.

The **acceleration** of the particle is given by (a_x, a_y, a_z) with

$$a_x = a_x(t) = \frac{d}{dt} v_x(t) \quad a_y = a_y(t) = \frac{d}{dt} v_y(t) \quad a_z = a_z(t) = \frac{d}{dt} v_z(t).$$

Here, (v_x, v_y, v_z) is the **velocity** of the particle at time $t > 0$:

$$v_x = v_x(t) = v_{0x} + \int_0^t a_x(t') dt' \quad v_y = v_y(t) = v_{0y} + \int_0^t a_y(t') dt' \quad \text{likewise for the z-axis.}$$

(v_{0x}, v_{0y}, v_{0z}) is the **velocity** of the particle at time $t = 0$.

Considering times t_1 and t_2 , with $t_2 > t_1$, we have

$$v_x(t_2) - v_x(t_1) = \int_{t_1}^{t_2} a_x(t) dt \quad v_y(t_2) - v_y(t_1) = \int_{t_1}^{t_2} a_y(t) dt \quad \text{likewise for the z-axis.}$$

2

Linear momentum and impulse

- A particle of mass m moving at **velocity** (v_x, v_y, v_z) has **linear momentum** (p_x, p_y, p_z) , with $p_x = mv_x$, $p_y = mv_y$, $p_z = mv_z$.
- From $v_x(t_2) - v_x(t_1) = \int_{t_1}^{t_2} a_x(t) dt$ $v_y(t_2) - v_y(t_1) = \int_{t_1}^{t_2} a_y(t) dt$ likewise for the z-axis, and multiplying through with mass m of the particle:
 $mv_x(t_2) - mv_x(t_1) = \int_{t_1}^{t_2} ma_x(t) dt$ $mv_y(t_2) - mv_y(t_1) = \int_{t_1}^{t_2} ma_y(t) dt$ likewise for the z-axis.
- From Newton's equation of motion,

$$\underbrace{p_x(t_2) - p_x(t_1)}_{\substack{\text{change in} \\ \text{linear momentum}}} = \underbrace{\int_{t_1}^{t_2} F_x(t) dt}_{\text{impulse}} \quad \underbrace{p_y(t_2) - p_y(t_1)}_{\substack{\text{change in} \\ \text{linear momentum}}} = \underbrace{\int_{t_1}^{t_2} F_y(t) dt}_{\text{impulse}} \quad \text{likewise for the z-axis.}$$

3

Linear momentum and impulse (cont.)

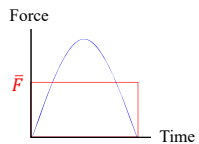
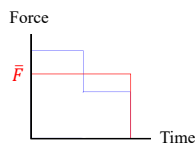
- If a particle is subject to a **force** (F_x, F_y, F_z) , its **linear momentum** changes:

$$\underbrace{p_x(t_2) - p_x(t_1)}_{\substack{\text{change in} \\ \text{linear momentum}}} = \underbrace{\int_{t_1}^{t_2} F_x(t) dt}_{\text{impulse}} \quad \underbrace{p_x(t_2) - p_x(t_1)}_{\substack{\text{change in} \\ \text{linear momentum}}} = \underbrace{\int_{t_1}^{t_2} F_x(t) dt}_{\text{impulse}} \quad \text{likewise for the z-axis.}$$

This is the **impulse-momentum theorem** (compare with the **work-energy theorem**): the **change in linear momentum** of a particle during a time interval equals the **impulse of the net force** that acts on the particle during that interval.

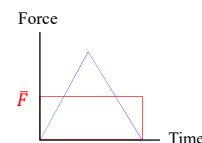
The *change* in momentum is called *impulse*!!

○ **Averaged Force** over time: $F_{\text{average}, x} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} F_x(t) dt,$



$$F_{\text{average}, y} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} F_y(t) dt.$$

$$F_{\text{average}, z} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} F_z(t) dt.$$



4

Problem. A movable notice-board of mass 4.0 kg is placed on a smooth floor. What is the **initial acceleration of the notice-board** when a horizontal stream of water, travelling at speed 8.0 m/s, strikes it at a rate of 1.0 kg/s for a duration of 52 s? The horizontal velocity of the water will become 0 m/s after it has struck the notice board.

Answer:

Let going right to be positive.

On the water,

$$F \times t = mv_f - mv_i$$

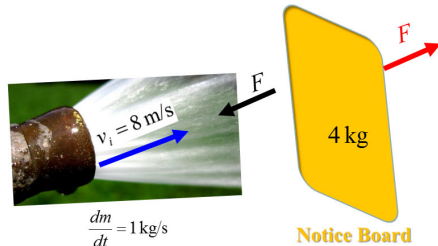
$$F = \frac{mv_f - mv_i}{t} = (v_f - v_i) \times \left(\frac{m}{t} \right)$$

$$= (0 - 8) \times 1 = -8 \text{ N}$$

(The water is receiving a force (directing to the left) that opposes it from going to the right.)

By the Newton's Third Law, the force exerted by the water will be 8 N pushing the notice board to the right. This force will accelerate the notice board. So,

$$F = ma \quad \rightarrow \quad a = \frac{F}{m} = \frac{8}{4} = 2 \text{ m/s}^2$$



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Linear momentum and impulse (cont.)

○ Notation.

$$J_x = \underbrace{\int_{t_1}^{t_2} F_x(t) dt}_{\text{impulse}}, \quad J_y = \underbrace{\int_{t_1}^{t_2} F_y(t) dt}_{\text{impulse}}, \quad J_z = \underbrace{\int_{t_1}^{t_2} F_z(t) dt}_{\text{impulse}}$$

Remark.

In terms of **linear momentum**, **Newton's equations of motion**:

$$F_x = ma_x = m \frac{dv_x}{dt} = \frac{d}{dt}(mv_x) = \frac{dp_x}{dt},$$

Newton's 2nd Law:

$$F_y = ma_y = m \frac{dv_y}{dt} = \frac{d}{dt}(mv_y) = \frac{dp_y}{dt},$$

Net force is equal to mass times acceleration.

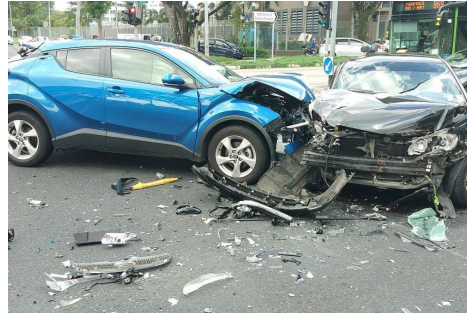
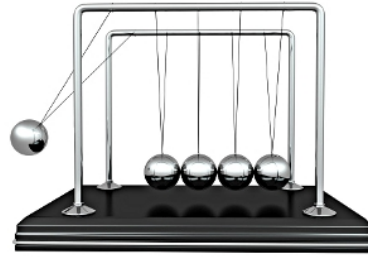
$$F_z = ma_z = m \frac{dv_z}{dt} = \frac{d}{dt}(mv_z) = \frac{dp_z}{dt}.$$

Net force is also equal to the changes of momentum in 1 second!!

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Collisions

A **collision** is any strong interaction between bodies that lasts a relatively short (and measurable) time – we can treat the bodies as an **isolated system**.



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Conservation of linear momentum



Consider an **isolated system of two particles**, A and B , that interact with nothing else but with each other, i.e., each particle exerts a force on the other.

On A :

$$\underbrace{p_{Ax}(t_2) - p_{Ax}(t_1)}_{\substack{\text{change in} \\ \text{linear momentum} \\ \text{of particle } A \text{ in } x\text{-direction}}} = \underbrace{\int_{t_1}^{t_2} F_{B \text{ on } A, x}(t) dt}_{\substack{\text{impulse on } A \\ \text{due to } B \text{ in } x\text{-direction}}}$$

$$\underbrace{p_{Ay}(t_2) - p_{Ay}(t_1)}_{\substack{\text{change in} \\ \text{linear momentum} \\ \text{of particle } A \text{ in } y\text{-direction}}} = \underbrace{\int_{t_1}^{t_2} F_{B \text{ on } A, y}(t) dt}_{\substack{\text{impulse on } A \\ \text{due to } B \text{ in } y\text{-direction}}}$$

likewise for the z -axis.

and on B :

$$\underbrace{p_{Bx}(t_2) - p_{Bx}(t_1)}_{\substack{\text{change in} \\ \text{linear momentum} \\ \text{of particle } B \text{ in } x\text{-direction}}} = \underbrace{\int_{t_1}^{t_2} F_{A \text{ on } B, x}(t) dt}_{\substack{\text{impulse on } B \\ \text{due to } A \text{ in } x\text{-direction}}}$$

$$\underbrace{p_{By}(t_2) - p_{By}(t_1)}_{\substack{\text{change in} \\ \text{linear momentum} \\ \text{of particle } B \text{ in } y\text{-direction}}} = \underbrace{\int_{t_1}^{t_2} F_{A \text{ on } B, y}(t) dt}_{\substack{\text{impulse on } B \\ \text{due to } A \text{ in } y\text{-direction}}}$$

likewise for the z -axis.

By the Newton's 3rd Law,

$$F_{A \text{ on } B, x}(t) = -F_{B \text{ on } A, x}(t), \quad F_{A \text{ on } B, y}(t) = -F_{B \text{ on } A, y}(t), \quad F_{A \text{ on } B, z}(t) = -F_{B \text{ on } A, z}(t).$$

Newton's 3rd Law

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Conservation of linear momentum (cont.)



Therefore,

$$\underbrace{p_{Ax}(t_2) - p_{Ax}(t_1)}_{\substack{\text{change in} \\ \text{linear momentum} \\ \text{of particle A in x-direction}}} = - \underbrace{[p_{Bx}(t_2) - p_{Bx}(t_1)]}_{\substack{\text{change in} \\ \text{linear momentum} \\ \text{of particle B in x-direction}}}$$

$$\underbrace{p_{Ay}(t_2) - p_{Ay}(t_1)}_{\substack{\text{change in} \\ \text{linear momentum} \\ \text{of particle A in y-direction}}} = - \underbrace{[p_{By}(t_2) - p_{By}(t_1)]}_{\substack{\text{change in} \\ \text{linear momentum} \\ \text{of particle B in y-direction}}}$$

... likewise for the z-axis.

Or,

$$\underbrace{p_{Ax}(t_2) + p_{Bx}(t_2)}_{\substack{\text{final total} \\ \text{linear momentum in x-direction}}} = \underbrace{p_{Ax}(t_1) + p_{Bx}(t_1)}_{\substack{\text{initial total} \\ \text{linear momentum in x-direction}}}$$

$$\underbrace{p_{Ay}(t_2) + p_{By}(t_2)}_{\substack{\text{final total} \\ \text{linear momentum in y-direction}}} = \underbrace{p_{Ay}(t_1) + p_{By}(t_1)}_{\substack{\text{initial total} \\ \text{linear momentum in y-direction}}}$$

... likewise for the z-axis.

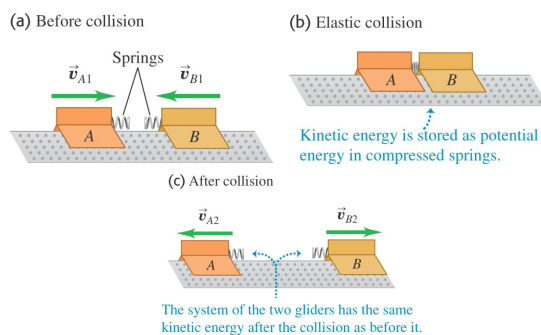
The total linear momentum of an isolated system of two interacting particles is constant – **conservation of linear momentum**. Note that the individual linear momenta of the particles that make up the system can change though.

9

Elastic and inelastic collisions

A **collision** in which the **total kinetic energy is conserved** is **elastic**.

Elastic collisions occur when the **forces** between the colliding bodies are **conservative**:



A **collision** in which the total **kinetic energy** after the collision is less than before the collision is **inelastic**.

An **inelastic collision** in which the colliding bodies stick together and move as one body after the collision is a **completely (perfectly) inelastic collision**:



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Example 1. Completely (Perfectly) **inelastic** collision in 1D



By **conservation of linear momentum**, $\overbrace{m_A v_{A1} + m_B v_{B1}}^{\text{Before Collision}} = \overbrace{(m_A + m_B) v_2}^{\text{After Collision}}$.

If $v_{B1} = 0$, $v_2 = \frac{m_A}{m_A + m_B} v_{A1}$.

$$\begin{aligned} \text{Total kinetic energy after collision } KE_2 &= \frac{1}{2} (m_A + m_B) v_2^2 = \frac{1}{2} (m_A + m_B) \left(\frac{m_A}{m_A + m_B} v_{A1} \right)^2 \\ &= \frac{m_A}{m_A + m_B} \left(\frac{1}{2} m_A v_{A1}^2 \right) = \left(\frac{m_A}{m_A + m_B} \right) KE_1 < KE_1 \end{aligned}$$

where KE_1 is the total kinetic energy before collision.

If $v_{B1} \neq 0$, we can also prove that the final total KE is smaller than initial total KE.

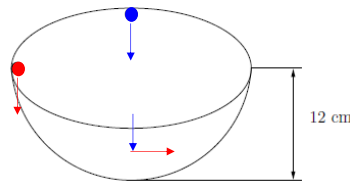
(I will leave it as your exercise!!)

Kinetic energy is always lost in a **completely (Perfectly) inelastic collision**.

This is the price to be paid for remaining together in the event of collision!!

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Problem. Two objects of masses **2.0 kg** and **6.0 kg** slide without friction down the sides of a hemispherical bowl as shown in the figure below. Both start with zero velocity at the lip, which is 12 cm above the bottom. The objects collide at the bottom of the bowl.



Just before the collision, they are moving at right angles to each other. To what maximum height above the bowl will the objects move if the collision is **completely inelastic**? You may model the objects as point particles.

Answer:

Both balls have the same height, hence they have the same speed when they reach the bottom. Let v be the velocity when they reach the bottom. By conservation of energy:

$$\frac{1}{2} m v^2 = m g h \rightarrow v = \sqrt{2 g h}$$

After the collision, let v' be their common velocity. By conservation of momentum in the absence of net external force,

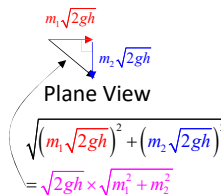
$$m_1 (\sqrt{2 g h} \rightarrow) + m_2 (\sqrt{2 g h} \downarrow) = (m_1 + m_2) v'$$

By Pythagorean Theorem: $v' = \frac{\sqrt{2 g h} \times \sqrt{m_1^2 + m_2^2}}{m_1 + m_2}$

But by conservation of **total energy**:

$$(m_1 + m_2) g h_{\max} = \frac{1}{2} (m_1 + m_2) v'^2$$

$$\begin{aligned} \rightarrow h_{\max} &= \frac{1}{2 g} v'^2 = \frac{1}{2 g} \left(\frac{2 g h (m_1^2 + m_2^2)}{(m_1 + m_2)^2} \right) = \frac{h (m_1^2 + m_2^2)}{(m_1 + m_2)^2} \\ &= \frac{0.12 (2^2 + 6^2)}{(2 + 6)^2} = 0.075 \text{ m} = \boxed{7.5 \text{ cm}} \end{aligned}$$



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Problem. A driver makes a reverse of his vehicle of mass 2100 kg (with driver) at 17 m/s in a sunny day, and has a rear-end collision with a stationary car whose mass is 1900 kg. Both cars lock bumpers and skid off together for a distance of 5.9 m with the wheels locked and finally crash into a coffeeshop so this accident becomes a court case. The driver mitigates that the road surface was smooth in order to reduce his responsibility. The judge is advised by experts that the legal standard for the coefficient of kinetic friction of road surface has to be at least 0.4 in a sunny day, and the camera footage shows that the collision was along a straight line. Is the mitigation of the driver valid?

Answer:

The total linear momentum of the two-car system is conserved because no net external force acts on the system during the collision. By conservation of linear momentum,

$$\underbrace{m_A v_{A1} + m_B v_{B1}}_{\text{Total momentum before collision}} = \underbrace{(m_A + m_B) v_{\text{common}}}_{\text{Total momentum after collision}}$$

$$\Rightarrow 2100 \times 17 = (2100 + 1900) \times v_{\text{common}} \Rightarrow v_{\text{common}} = 8.925 \text{ m/s}$$

The kinetic energy of the 2 cars subsequently dissipates as heat due to the kinetic friction of the road surface.

$$\text{So } \frac{1}{2} (m_1 + m_2) v_{\text{common}}^2 = \underbrace{(\mu_k (m_1 + m_2) g)}_{\text{kinetic friction on road surface}} \times d$$

$$\Rightarrow \frac{1}{2} v_{\text{common}}^2 = \mu_k g \times d$$

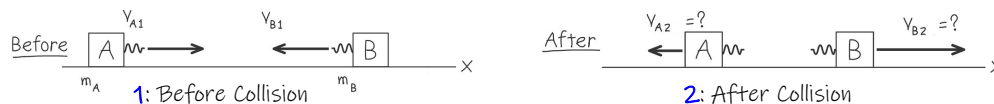
$$\Rightarrow \frac{8.925^2}{2} = \mu_k \times 9.8 \times 5.9$$

$$\Rightarrow \mu_k = 0.6888 > 0.4$$

The mitigation of the driver is **not** valid.

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Example 2. Elastic collision in 1D



By **conservation of linear momentum**, $\overbrace{m_A v_{A1} + m_B v_{B1}}^{\text{Before Collision}} = \overbrace{m_A v_{A2} + m_B v_{B2}}^{\text{After Collision}}$.

$$\Rightarrow m_A (v_{A1} - v_{A2}) = m_B (-v_{B1} + v_{B2}) \dots (1)$$

By **conservation of kinetic energy**, $\frac{1}{2} m_A v_{A1}^2 + \frac{1}{2} m_B v_{B1}^2 = \frac{1}{2} m_A v_{A2}^2 + \frac{1}{2} m_B v_{B2}^2$

$$\Rightarrow m_A (v_{A1}^2 - v_{A2}^2) = -m_B (v_{B1}^2 - v_{B2}^2)$$

$$\Rightarrow m_A (v_{A1} + v_{A2})(v_{A1} - v_{A2}) = -m_B (v_{B1} + v_{B2})(v_{B1} - v_{B2}) \dots (2)$$

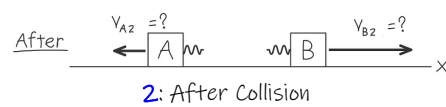
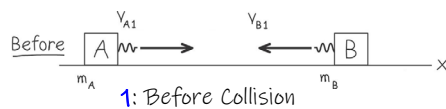
$$(2)/(1) \Rightarrow (v_{A1} + v_{A2}) = (v_{B1} + v_{B2})$$

Systematical scripts for relative velocity equation for elastic collision??

$$\Rightarrow v_{A1} - v_{B1} = -(v_{A2} - v_{B2})$$

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Example 2. Elastic collision in 1D (cont.)



By **conservation of linear momentum**,

$$\overbrace{m_A v_{A1} + m_B v_{B1}}^{\text{Before Collision}} = \overbrace{m_A v_{A2} + m_B v_{B2}}^{\text{After Collision}}$$

If $v_{B1} = 0$, $m_A v_{A1} = m_A v_{A2} + m_B v_{B2} \dots (1)$

By **conservation of kinetic energy** (previous slide),

$$v_{A1} - \cancel{v_{B1}^0} = -(v_{A2} - v_{B2})$$

$$\rightarrow v_{A2} = -v_{A1} + v_{B2} \dots (2)$$

Sub (2) to (1):

$$m_A v_{A1} = m_A (-v_{A1} + v_{B2}) + m_B v_{B2}$$

$$\rightarrow 2m_A v_{A1} = m_A v_{B2} + m_B v_{B2}$$



$$2m_A v_{A1} = v_{B2} (m_A + m_B)$$

$$\rightarrow v_{B2} = \frac{2m_A}{m_A + m_B} v_{A1} \quad \text{Assume } v_{B1} = 0$$

From (2):

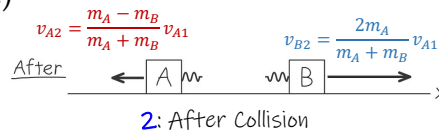
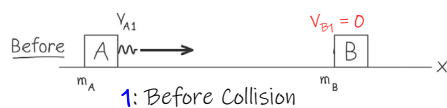
$$\begin{aligned} v_{A2} &= -v_{A1} + \frac{2m_A}{m_A + m_B} v_{A1} \\ &= \frac{-m_A - m_B + 2m_A}{m_A + m_B} v_{A1} \end{aligned}$$

$$\rightarrow v_{A2} = \frac{m_A - m_B}{m_A + m_B} v_{A1} \quad \text{Assume } v_{B1} = 0$$

If object B was initially not moving, after collision object A can have 3 options. What are these options?

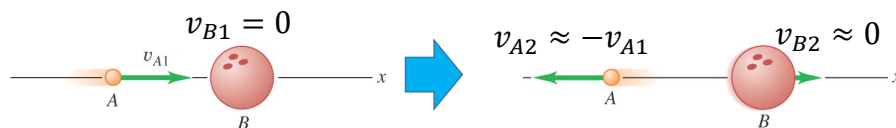
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Example 2. Elastic collision in 1D (cont.)

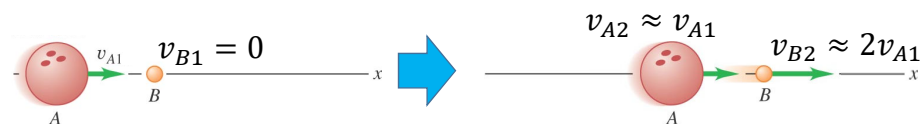


Remarks.

- For $m_A \ll m_B$ (e.g., $m_A = 1$, $m_B = 10000$)



- For $m_A \gg m_B$ (e.g., $m_A = 10000$, $m_B = 1$)



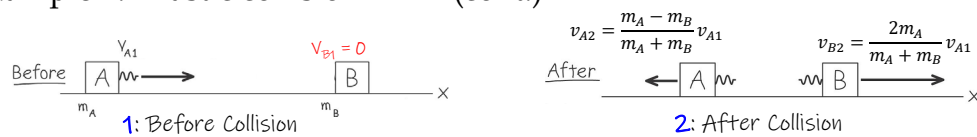
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The ratio of m_A versus m_B in elastic collision



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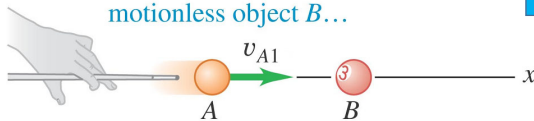
Example 2. Elastic collision in 1D (cont.)



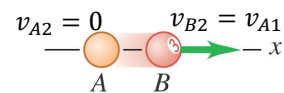
Remarks (cont.).

- For $m_A = m_B$

When a moving object A has a 1-D elastic collision with an equal-mass, motionless object B...

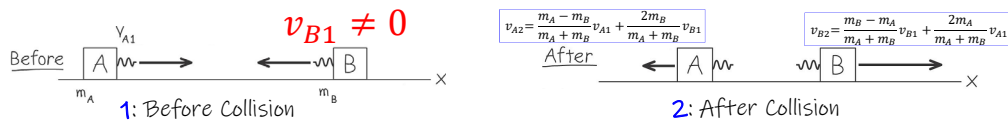


...all of A's momentum and kinetic energy are transferred to B.



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Example 3. Elastic collision in 1D (cont.)



By **conservation of linear momentum**, $\overbrace{m_A v_{A1} + m_B v_{B1}}^{\text{Before Collision}} = \overbrace{m_A v_{A2} + m_B v_{B2}}^{\text{After Collision}} \dots (1)$

By **conservation of kinetic energy** (previous slide),

$$v_{A1} - v_{B1} = -(v_{A2} - v_{B2}) \quad \rightarrow \quad v_{A2} = v_{B2} + v_{B1} - v_{A1} \dots (2)$$

Sub (2) to (1): $m_A v_{A1} + m_B v_{B1} = m_A (v_{B2} + v_{B1} - v_{A1}) + m_B v_{B2}$

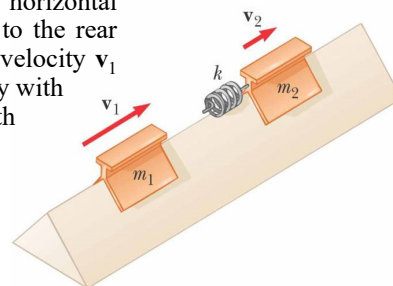
$$(m_B - m_A) v_{B1} + 2m_A v_{A1} = (m_A + m_B) v_{B2}$$

$$\rightarrow v_{B2} = \frac{m_B - m_A}{m_A + m_B} v_{B1} + \frac{2m_A}{m_A + m_B} v_{A1}$$

Similarly: $v_{A2} = \frac{m_A - m_B}{m_A + m_B} v_{A1} + \frac{2m_B}{m_A + m_B} v_{B1}$

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Problem. Two gliders are set in motion on a float horizontal air track. A spring of force constant k is attached to the rear side of one glider. The first glider of mass m_1 has velocity v_1 and the second glider of mass m_2 moves more slowly with velocity v_2 as shown in figure. When m_1 collides with the spring attached to m_2 and compresses the spring to its maximum compression $x_{\max}$, the velocity of the gliders is v . In terms of v_1 , v_2 , m_1 , m_2 , and k , find



(i) the velocity v at maximum compression,

Answer:

Initially $v_1 > v_2$ so m_1 will catch up with m_2 . When the compression of spring is going on, m_1 will be getting closer to m_2 so v_1 is still faster than v_2 . When the compression is maximum (i.e., m_1 cannot compress the spring further), both m_1 and m_2 will move with the same velocity at that very short instant. Subsequently, decompression starts and m_1 and m_2 will move with different velocities again.

When the spring is fully compressed, each cart moves with same velocity v at that instant. Apply conservation of momentum for the system of two gliders,

$$m_1 v_1 + m_2 v_2 = (m_1 + m_2) v \quad \rightarrow \quad v = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$$

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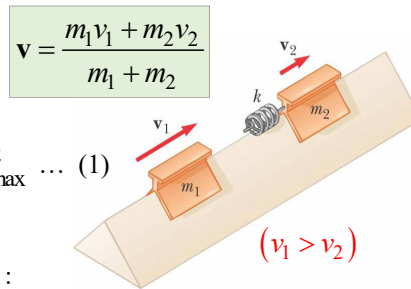
(ii) the maximum compression $x_{\max}$

Answer:

By conservation of energy,

$$\frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 = \frac{1}{2}(m_1 + m_2)v^2 + \frac{1}{2}kx_{\max}^2 \quad \dots (1)$$

Substitute $v = \frac{m_1v_1 + m_2v_2}{m_1 + m_2}$ from part (i) to eqn (1) :



$$m_1v_1^2 + m_2v_2^2 = (m_1 + m_2) \left(\frac{m_1v_1 + m_2v_2}{m_1 + m_2} \right)^2 + kx_{\max}^2 \Rightarrow m_1v_1^2 + m_2v_2^2 = \frac{m_1^2v_1^2 + 2m_1v_1m_2v_2 + m_2^2v_2^2}{m_1 + m_2} + kx_{\max}^2$$

$$\Rightarrow m_1^2v_1^2 + m_1m_2v_2^2 + m_2m_1v_1^2 + m_2^2v_2^2 = (m_1^2v_1^2 + 2m_1v_1m_2v_2 + m_2^2v_2^2) + k(m_1 + m_2)x_{\max}^2$$

$$\Rightarrow m_1m_2v_2^2 - 2m_1v_1m_2v_2 + m_2m_1v_1^2 = k(m_1 + m_2)x_{\max}^2 \Rightarrow m_1m_2(v_2 - v_1)^2 = k(m_1 + m_2)x_{\max}^2$$

$$\Rightarrow x_{\max}^2 = \frac{m_1m_2(v_2 - v_1)^2}{k(m_1 + m_2)} \Rightarrow x_{\max} = -\frac{(v_2 - v_1)}{\sqrt{\frac{k(m_1 + m_2)}{m_1m_2}}} = (v_1 - v_2) \sqrt{\frac{m_1m_2}{k(m_1 + m_2)}}$$

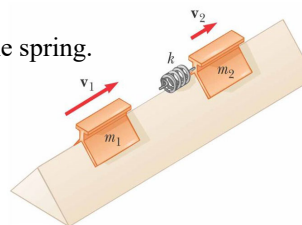
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(iii) the velocity of each glider after m_1 has lost contact with the spring.

Answer:

By conservation of momentum:

$$m_1v_1 + m_2v_2 = m_1v_{1f} + m_2v_{2f} \quad \dots (1)$$



$$\text{By conservation of energy: } \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 = \frac{1}{2}m_1v_{1f}^2 + \frac{1}{2}m_2v_{2f}^2 \quad \dots (2)$$

This is same as elastic collision where the kinetic energy is conserved except that a partial KE of both gliders was transferred to the spring, and later totally returned to the gliders. Therefore

After separation:

$$v_{A2} = \frac{m_A - m_B}{m_A + m_B} v_{A1} + \frac{2m_B}{m_A + m_B} v_{B1}$$

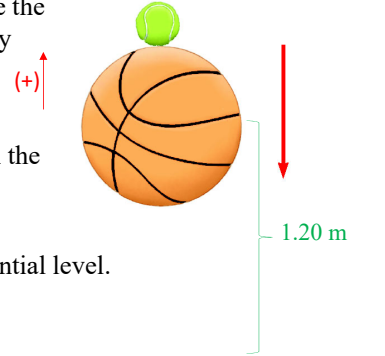
$$v_{B2} = \frac{m_B - m_A}{m_A + m_B} v_{B1} + \frac{2m_A}{m_A + m_B} v_{A1}$$

$$v_{m_1} = \frac{m_1 - m_2}{m_1 + m_2} v_1 + \frac{2m_2}{m_1 + m_2} v_2$$

$$v_{m_2} = \frac{m_2 - m_1}{m_1 + m_2} v_2 + \frac{2m_1}{m_1 + m_2} v_1$$

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Question. A tennis ball of mass 57.0 g is held just above a basketball of mass 590 g. With their centers vertically aligned, both balls are released from rest at the same time to fall through a distance of 1.20 m as shown in the figure. Assume that an elastic collision with the ground instantaneously reverses the velocity of the basketball while the tennis ball is still moving down, and the two balls meet immediately in an elastic collision. Both balls can be treated as two separate point masses.



- (i) Find the magnitude of the downward velocity with which the basketball reaches the ground.

Answer:

Take going **up** as positive, and the ground as the zero potential level.

$$\begin{aligned} v_y^2 &= v_{0y}^2 + 2a_{0y}(y - y_0) \\ &= 0^2 + 2(-9.80 \text{ m/s}^2)(-1.20 \text{ m}) \end{aligned}$$

→ $v_y = \boxed{-4.85 \text{ m/s}}$ **Magnitude** of the downward velocity with which the basketball reaches the ground is 4.85 m/s.

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(ii) To what height does the tennis ball rebound?

Answer:

Label tennis ball as A, and basket ball as B. Let the subscript 1 indicates before collision, and subscript 2 after collision. Take going up as positive. Immediately after the basketball rebounds from the floor, the basket ball will meet the tennis ball in an elastic collision. The velocities of the two ball just before the collision are:

For the tennis ball (still going down): $v_{A1} = v = -4.85 \text{ m/s}$

For the basketball (already bounced and is going up): $v_{B2} = -v_y = -(-4.85) = +4.85 \text{ m/s}$

By conservation of momentum:

$$(57.0 \text{ g})(-4.85 \text{ m/s}) + (590 \text{ g})(+4.85 \text{ m/s}) = (57.0 \text{ g})v_{A2} + (590 \text{ g})v_{B2}$$

$$\rightarrow v_{B2} = 4.38 - 0.0966v_{A2} \dots (1)$$

By relative velocity of elastic collision: $v_{A1} - v_{B1} = -(v_{A2} - v_{B2})$

$$\rightarrow v_{A2} - v_{B2} = -(v_{A1} - v_{B1}) = -(-4.85 - (+4.85)) = 9.7 \dots (2)$$

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$$\begin{cases} v_{B2} = 4.38 - 0.0966v_{A2} \dots (1) \\ v_{A2} - v_{B2} = 9.7 \dots (2) \end{cases}$$

$$\rightarrow v_{A2} - (4.38 - 0.0966v_{A2}) = 9.70$$

$$\rightarrow (1 + 0.0966)v_{A2} = 9.70 + 4.38$$

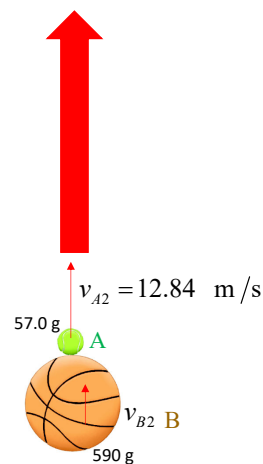
$$\rightarrow v_{A2} = \frac{14.08}{1.0966} = +12.84 \text{ m/s}$$

The vertical displacement of the tennis ball during its rebound is given by

$$v_y^2 = v_{0y}^2 + 2a_{0y}(y - y_0)$$

$$0^2 = 12.84^2 + 2(-9.8)(\Delta y)$$

$$\rightarrow \Delta y = \frac{0^2 - 12.84^2}{2(-9.80)} = 8.41 \text{ m} > 1.2 \text{ m}$$

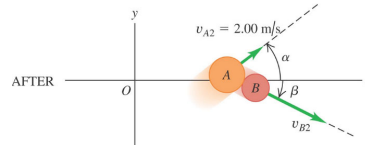
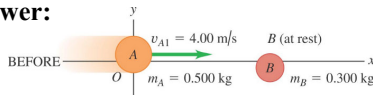


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2-Directional Collision

Problem. The figure shows an elastic collision of two pucks on a frictionless air-hockey ring, where puck A has an initial velocity of 4.00 m/s in the positive x -direction and a final velocity of 2.00 m/s in an unknown direction α , and puck B is initially at rest. Find the final speed v_{B2} of puck B and the angles α and β .

Answer:



As the collision is not aligned, there is a deflection. By **conservation of linear momentum**,

$$\text{Horizontally: } m_A v_{A1} + m_B v_{B1} = m_A v_{A2} \cos \alpha + m_B v_{B2} \cos \beta \rightarrow 2 = \cos \alpha + 0.3 v_{B2} \cos \beta \dots (1)$$

$$\text{Vertically: } 0 = m_A v_{A2} \sin \alpha - m_B v_{B2} \sin \beta \rightarrow 0 = \sin \alpha - 0.3 v_{B2} \sin \beta \dots (2)$$

$$\text{By conservation of kinetic energy, } \frac{1}{2} m_A v_{A1}^2 = \frac{1}{2} m_A v_{A2}^2 + \frac{1}{2} m_B v_{B2}^2$$

$$\rightarrow 4 = 1 + 0.15 v_{B2}^2 \rightarrow v_{B2} = \sqrt{\frac{3}{0.15}} = \sqrt{20} \text{ m/s} = \boxed{4.47 \text{ m/s}} \dots (3)$$

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$$\text{From (2): } 0 = \sin \alpha - 0.3 v_{B2} \sin \beta$$

$$\rightarrow \sin \alpha = 1.34163 \sin \beta$$

$$\rightarrow \sin^2 \alpha = 1.8 \sin^2 \beta \dots (3)$$

$$\text{From (1): } 2 = \cos \alpha + 0.3 v_{B2} \cos \beta$$

$$\rightarrow \cos \alpha = 2 - 0.3 \sqrt{20} \cos \beta$$

$$\cos^2 \alpha = (2 - 0.3 \sqrt{20} \cos \beta)^2$$

$$1 - \sin^2 \alpha = 4 - 1.2 \sqrt{20} \cos \beta + 1.8 \cos^2 \beta$$

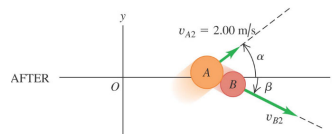
$$\text{Use (3)} \rightarrow 1 - 1.8 \sin^2 \beta = 4 - 1.2 \sqrt{20} \cos \beta + 1.8 \cos^2 \beta$$

$$\rightarrow 1 = 4 - 1.2 \sqrt{20} \cos \beta + 1.8 \cos^2 \beta + 1.8 \sin^2 \beta$$

$$\rightarrow 1 = 4 - 1.2 \sqrt{20} \cos \beta + 1.8$$

$$\rightarrow \cos \beta = \frac{4 + 1.8 - 1}{1.2 \sqrt{20}} = \frac{4.8}{5.3665} = 0.8944$$

$$\rightarrow \beta = \boxed{26.57^\circ}$$



$$2 = \cos \alpha + 0.3 v_{B2} \cos \beta \dots (1)$$

$$0 = \sin \alpha - 0.3 v_{B2} \sin \beta \dots (2)$$

$$v_{B2} = 4.47 \text{ m/s} \dots (3)$$

The rest is 100% algebra in trigonometry functions!!

$$\text{From (3): } \sin^2 \alpha = 1.8 \sin^2 \beta$$

$$\rightarrow \sin \alpha = \sqrt{1.8 \sin^2 26.57^\circ} = \sqrt{0.3601} = 0.6001$$

$$\rightarrow \alpha = \boxed{36.88^\circ}$$

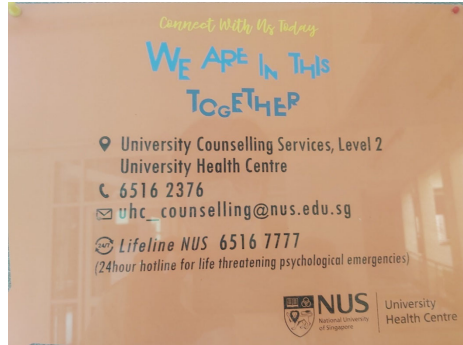
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Helplines to de-stress

We are not superman/supergirl. Sometime and somewhere in our life journey we will need help. Here they are:

Lifeline NUS (for life threatening psychological emergencies):
6516 7777 (24 hours)

6516 2376 (Office hours)
uhc_counselling@nus.edu.sg



Samaritans of Singapore Hotline: 1800 221 4444
Institute of Mental Health's Helpline: 6389 2222
Singapore Association of Mental Health Helpline: 1800 283 7019